Reviewer Pack — Standard Young Tableau Count Exponential Gap in Symmetric Po…

Entry 04216a9a3c0d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-10 09:21:18 UTC

Standard Young Tableau Count Exponential Gap in Symmetric Power Decompositions

Entry ID: `04216a9a3c0d` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-10 09:21:18 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Combinatorics (Standard Young Tableaux) **Field B** (complexity object): Monotone Circuit Size for Permanent

Statement:

> For every $n \geq 2$, the number of standard Young tableaux of shape $\lambda = (n-1,1)$ for the permanent polynomial exceeds the number for the determinant polynomial by a factor of $\Omega(2^n)$.

Rationale (proposer's reasoning):

> The shape $\lambda = (n-1,1)$ corresponds to the dominant weight of the symmetric power representation, which captures the combinatorial structure of monomial expansions. The exponential gap in tableau counts reflects the inherent complexity of symmetric vs alternating decompositions in monotone circuits.

Taxonomy category: `GCT_DET_PERM` (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `0ea97f676b97887e`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> default: >=80% seeds must support, no counterexample

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.95	SAFE	UNCERTAIN

Barrier	Verdict	Confidence	LLM1	LLM2
ALGEBRIZATION	SAFE	0.95	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.95	UNCERTAIN	SAFE

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def hook_length_formula(n):
        return math.factorial(n + 1) // (math.prod(range(1, n)) * math.prod(range(2, n + 2)))

    n = random.randint(5, 40)
    lambda_permanent = (n - 1, 1)
    lambda_determinant = (n, 1)

    T_permanent = hook_length_formula(n)
    T_determinant = hook_length_formula(n)

    ratio = math.log2(T_permanent / T_determinant)
    conjecture_holds = ratio > n / 2
    counterexample = "" if conjecture_holds else f"Ratio {ratio} <= {n/2}"

    return {
        "metric_name": "log2(T_permanent / T_determinant)",
        "metric_value": ratio,
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or list(range(2, 30)) + [101, 103, 107, 109, 113]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_ratio = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)
```

```python
    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_ratio} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_ratio} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, r in zip(seeds, results) if not r["conjecture_hol
        print(f"RESULT: FALSIFIED counterexample=\"Ratio {r['metric_value']} <= {n/2}\" first_fail
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11ee74bb.py", line 48, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11ee74bb.py", line 31, in run_trial
    ratio = math.log2(T_permanent / T_determinant)
                      ~~~~~~~~~~~~^~~~~~~~~~~~~~~
ZeroDivisionError: division by zero
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> Test crashed with ZeroDivisionError, preventing result evaluation. No counterexample found but support fraction cannot be assessed. | next: Fix division-by-zero handling in test code and re-run with diverse seeds

11. Audit log (LLM calls)

Total LLM calls: 12

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	109880
2	propose	ollama_remote	qwen3:8b	0	0	98816
3	propose	ollama_remote	qwen3:8b	0	0	112183
4	propose	ollama_remote	qwen3:8b	0	0	35199
5	propose	ollama_remote	qwen3:8b	0	0	111532
6	propose	ollama_remote	qwen3:8b	0	0	96052
7	preregistration	ollama_remote	qwen3:8b	0	0	24111
8	novelty	ollama_remote	qwen3:8b	0	0	20674

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
9	novelty	ollama_remote	qwen3:8b	0	0	17173
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10214
11	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	5984
12	judge	ollama_remote	qwen3:8b	0	0	15804

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 657621 ms total latency. Provider mix: {'ollama_remote': 12}

(full prompt+response transcripts available in `research/audit/04216a9a3c0d.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/04216a9a3c0d.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/04216a9a3c0d.tar.gz` (if generated)